

Intensive Property

Definition. A physical property P of a thermodynamic system is called *intensive* if its value does not depend on the extent (size or mass) of the system. Formally, suppose a homogeneous system is partitioned into two subsystems with masses m_1 and m_2 , each in the same thermodynamic state as the whole. If P satisfies

$$P(\lambda m) = P(m), \quad \lambda > 0,$$

then P is intensive. This contrasts with an *extensive* property E , which scales linearly with system size, $E(\lambda m) = \lambda E(m)$. Common intensive properties include temperature T , pressure p , density ρ , and specific quantities such as molar volume $\bar{v} = V/n$.

Worked example. Consider a container of water at uniform temperature $T = 25^\circ\text{C}$ and density $\rho = 997 \text{ kg/m}^3$. If the container is divided into two equal halves, each half still has $T = 25^\circ\text{C}$ and $\rho = 997 \text{ kg/m}^3$ – these values are unchanged by halving the mass. In contrast, the total volume V and total mass m are each halved, confirming that V and m are extensive while T and ρ are intensive.

Key theorem. If E is an extensive property that is a homogeneous function of degree 1 in the extensive variables $\{n_i\}$ (mole numbers) at fixed intensive variables, Euler's theorem gives

$$E(n_1, \dots, n_k) = \sum_{i=1}^k n_i \left(\frac{\partial E}{\partial n_i} \right)_{T, p, n_{j \neq i}},$$

where each partial molar derivative $\bar{e}_i = (\partial E / \partial n_i)$ is itself intensive. This identity underlies the definition of chemical potential and partial molar quantities in chemical thermodynamics.

Lab cell (SymPy).

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import sympy as sp

n, V_m = sp.symbols('n V_m', positive=True) # V_m: molar volume (in
V = n * V_m                                # total volume (extens
```

```
lam = sp.symbols('lambda', positive=True)
scaled = V.subs(n, lam*n)
print(sp.simplify(scaled / V))          # -> lambda, confirms
print(sp.simplify(V_m.subs(n, lam*n) / V_m))  # -> 1, confirms V_m
```